

Theorem 3.2.3 *A system $(X, T, \mu,)$, where X is a Riemannian manifold and T is a diffeomorphism, is hyperbolic iff for almost all $x \in X$*

$$\lambda(x, v) \neq 0 \quad \forall v \in \mathcal{T}_x X, \quad v \neq 0.$$

PROOF. Keep in mind that the L.E. are defined a.e. by Theorem 3.2.2 if the system is hyperbolic, then all the LE are different from zero. The other implication is almost as trivial. Define $E^s(x) = \{v \in \mathcal{T}_x \mid \lambda(x, v) < 0\}$; then consider the Dynamical system (X, T^{-1}, μ) and its LE $\lambda^-(x, v)$ and define $E^u(x) = \{v \in \mathcal{T}_x \mid \lambda^- < 0\}$. Next, let

$$\rho(x) = \sup\{\lambda(x, v), \lambda_-(x, w) \mid v \in E^s(x), w \in E^u(x)\}$$

clearly $\rho(x) < 0$ a.e.. Then setting $\nu(x) = e^{-\rho(x)/2}$ and

$$c(x) = \sup\{\nu(x)^n \|D_x T^n v\|, \nu(x)^n \|D_x T^{-n} w\| \mid v \in E^s(x); w \in E^u(x)\}_{n \in \mathbb{N}},$$

the theorem is proven. $\square$

3.2.1 Examples

Logistic map

do by conjugation and by showing via a dyn. proof that $\lambda(x) > 0$ for each $x \notin \cup_{n \in \mathbb{N}} T^{-n}(\{0\})$.

The above facts show that the hyperbolic systems have an infinitesimal behavior similar to our toy model. Yet, in the arguments of the previous chapter it was essential the existence of local manifolds (lines). In the following sections we will see that hyperbolic systems do have local manifolds as well.

3.3 Invariant manifold of a fixed point

In order to gain a feeling for the type of results we will present in the next section let us consider the simplest possible case in which the existence of invariant manifolds arises: the Hadamard-Perron theorem.

Definition 3.3.1 *Given a smooth map $T : X \rightarrow X$, X being a Riemannian manifold, and a fixed point $p \in X$ (i.e. $Tp = p$) we call (local) stable manifold (of size δ) a manifold $W^s(p)$ such that⁴*

$$W^s(p) = \{x \in B_\delta(x) \subset X \mid \lim_{n \rightarrow \infty} d(T^n x, p) = 0\}.$$

⁴Sometime we will write $W_\delta^s(p)$ when the size really matters. By $B_\delta(x)$ we will always mean the open ball of radius δ centered at x .

Analogously, we will call (local) unstable manifold (of size δ) a manifold $W^u(p)$ such that

$$W^u(p) = \{x \in B_\delta(x) \subset X \mid \lim_{n \rightarrow \infty} d(T^{-n}x, p) = 0\}.$$

It is quite clear that $TW^s(p) \subset W^s(p)$ and $TW^u(p) \supset W^u(p)$ (Problem 3.7). Less clear is that these sets deserve the name “manifold.” Yet, if one thinks of the Arnold cat at the point zero (which is a fixed point) it is obvious that the stable and unstable manifolds at zero are just segments in the stable and unstable direction, the next Theorem shows that this is a quite general situation.

Theorem 3.3.2 (Hadamard-Perron) *Consider an invertible map $T : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T \in \mathcal{C}^{(1)}(U, \mathbb{R}^2)$, such that $T0 = 0$ and*

$$D_0T = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix} \quad (3.3.1)$$

where $0 < \mu < 1 < \lambda$.⁵ That is, the map T is hyperbolic at the fixed point 0. Then there exists stable and unstable manifolds at 0. Moreover, $T_0W^{s(u)}(0) = E^{s(u)}(0)$ where $E^{s(u)}(0)$ are the expanding and contracting subspaces of D_0T .

PROOF. We will deal explicitly only with the unstable manifold since the stable one can be treated exactly in the same way by considering T^{-1} instead of T .

Since the map is continuously differentiable for each $\varepsilon > 0$ we can choose $\delta > 0$ so that, in a 2δ -neighborhood of zero, we can write

$$T(x) = D_0Tx + R(x) \quad (3.3.2)$$

where $\|R(x)\| \leq \varepsilon\|x\|$, $\|D_xR\| \leq \varepsilon$.

Existence—a fixed point argument

The first step is to decide how to represent manifolds. In the present case, since we deal only with curves, it seems very reasonable to consider the set of curves $\Gamma_{\delta,c}$ passing through zero and “close” to being horizontal, that is the differentiable functions $\gamma : [-\delta, \delta] \rightarrow \mathbb{R}^2$ of the form

$$\gamma(t) = \begin{pmatrix} t \\ u(t) \end{pmatrix}$$

⁵Notice that if D_0T has eigenvalues $0 < \mu < 1 < \lambda$ then one can always perform a change of variables such that (3.3.1) holds.

and such that $\gamma(0) = 0$; $\|(1, 0) - \gamma'\|_\infty \leq c$. It is immediately clear that any smooth curve passing through zero and with tangent vector, at each point, in the cone $\mathcal{C} := \{(a, b) \in \mathbb{R}^2 \mid |\frac{b}{a}| \leq c\}$, can be associated to a unique element of $\Gamma_{\delta, c}$, just consider the part of the curve contained in the strip $\{(x, y) \in \mathbb{R}^2 \mid |x| \leq \delta\}$. Moreover, if $\gamma \in \Gamma_{\delta, c}$ then $\gamma \subset B_{2\delta}(0)$, provided $c \leq 1/2$.

Notice that it suffice to specify the function u in order to identify uniquely an element in $\Gamma_{\delta, c}$. It is then natural to study the evolution of a curve through the change in the associated function.

To this end let us investigate how the image of a curve in $\Gamma_{\delta, c}$ under T looks like.

$$T\gamma(t) = \begin{pmatrix} \lambda t + R_1(t, u(t)) \\ \mu u(t) + R_2(t, u(t)) \end{pmatrix} := \begin{pmatrix} \alpha_u(t) \\ \beta_u(t) \end{pmatrix}.$$

At this point the problem is clearly that the image it is not expressed in the way we have chosen to represent curves, yet this is easily fixed. First of all, $\alpha_u(0) = \beta_u(0) = 0$. Second, by choosing $\varepsilon < \lambda$, we have $\alpha'_u(t) > 0$, that is, α_u is invertible. In addition, $\alpha_u([-\delta, \delta]) \supset [-\lambda\delta + \varepsilon\delta, \lambda\delta - \varepsilon\delta] \supset [-\delta, \delta]$, provided $\varepsilon \leq \lambda^{-1}$. Hence, α_u^{-1} is a well defined function from $[-\delta, \delta]$ to itself. Finally,

$$\left| \frac{d}{dt} \beta_u \circ \alpha_u^{-1}(t) \right| = \left| \frac{\beta'_u(\alpha_u^{-1}(t))}{\alpha'_u(\alpha_u^{-1}(t))} \right| \leq \frac{\mu c + \varepsilon}{\lambda - \varepsilon} \leq c$$

where, again, we have chosen $\varepsilon \leq \frac{c(\lambda - \mu)}{1 + c}$.

We can then consider the map $\tilde{T} : \Gamma_{\delta, c} \rightarrow \Gamma_{\delta, c}$ defined by

$$\tilde{T}\gamma(t) := \begin{pmatrix} t \\ \beta_u \circ \alpha_u^{-1}(t) \end{pmatrix} \quad (3.3.3)$$

which associates to a curve in $\Gamma_{\delta, c}$ its image under T written in the chosen representation. It is now natural to consider the set of functions $B_{\delta, c} = \{u \in \mathcal{C}^{(1)}([-\delta, \delta]) \mid u(0) = 0, |u'|_\infty \leq c\}$ in the vector space $Lip([-\delta, \delta])$.⁶ As we already noticed $B_{\delta, c}$ is in one-one correspondence with $\Gamma_{\delta, c}$, we can thus consider the operator $\hat{T} : Lip([-\delta, \delta]) \rightarrow Lip([-\delta, \delta])$ defined by

$$\hat{T}u = \beta_u \circ \alpha_u^{-1} \quad (3.3.4)$$

From the above analysis follows that $\hat{T}(B_{\delta, c}) \subset B_{\delta, c}$ and that $\hat{T}u$ determines uniquely the image curve.

The problem is then reduced to studying the map $\hat{T}$. The easiest, although probably not the most productive, point of view is to show that $\hat{T}$ is a contraction in the sup norm. Note that this creates a little problem since $\mathcal{C}^{(1)}$ it

⁶This are the Lipschitz functions on $[-\delta, \delta]$, that is the functions such that $\sup_{t, s \in [-\delta, \delta]} \frac{|u(s) - u(t)|}{|t - s|} < \infty$.

is not closed in the sup norm (and not even $Lip([-δ, δ])$ is closed). Yet, the set $B_{δ,c}^* = \{u \in Lip([-δ, δ]) \mid u(0) = 0, \sup_{t,s \in [-δ, δ]} \frac{|u(s)-u(t)|}{|t-s|} < c\}$ is closed (see Problem 3.8). Thus $\overline{B_{δ,c}} \subset B_{δ,c}^*$. This means that, if we can prove that the sup norm is contracting, then the fixed point will belong to $B_{δ,c}^*$ and we will obtain only a Lipschitz curve. We will need a separate argument to prove that the curve is indeed smooth.

Let us start to verify the contraction property. Notice that

$$\alpha_u^{-1}(t) = \lambda^{-1}t + \lambda^{-1}R_1(\alpha_u^{-1}(t), u(\alpha_u^{-1}(t))),$$

thus, given $u_1, u_2 \in B_{δ,c}$, by Lagrange Theorem

$$\begin{aligned} |\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| &\leq \lambda^{-1} |\langle \nabla_\zeta R_1, (\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t), u_1(\alpha_{u_1}^{-1}(t)) - u_2(\alpha_{u_2}^{-1}(t))) \rangle| \\ &\leq \frac{\varepsilon}{\lambda} \{ |\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| + |u_1(\alpha_{u_2}^{-1}(t)) - u_2(\alpha_{u_2}^{-1}(t))| \}. \end{aligned}$$

This implies immediately

$$|\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| \leq \frac{\lambda^{-1}\varepsilon}{1 - \lambda^{-1}\varepsilon} \|u_1 - u_2\|_\infty. \quad (3.3.5)$$

On the other hand

$$\begin{aligned} |\beta_{u_1}(t) - \beta_{u_2}(t)| &\leq \mu |u_1(t) - u_2(t)| + |\langle \nabla_\zeta R_2, (0, u_1(t) - u_2(t)) \rangle| \\ &\leq (\mu + \varepsilon) \|u_1 - u_2\|_\infty. \end{aligned} \quad (3.3.6)$$

Moreover,

$$|\beta'_u(t)| \leq \mu + \varepsilon. \quad (3.3.7)$$

Collecting the estimates (3.3.5, 3.3.6, 3.3.7) readily yields

$$\begin{aligned} \|\hat{T}u_1 - \hat{T}u_2\|_\infty &\leq \|\beta_{u_1} \circ \alpha_{u_1}^{-1} - \beta_{u_1} \circ \alpha_{u_2}^{-1}\|_\infty + \|\beta_{u_1} \circ \alpha_{u_2}^{-1} - \beta_{u_2} \circ \alpha_{u_2}^{-1}\|_\infty \\ &\leq \left\{ [\mu + \varepsilon] \frac{\lambda^{-1}\varepsilon}{1 - \lambda^{-1}\varepsilon} + (\mu + \varepsilon) \right\} \|u_1 - u_2\|_\infty \\ &\leq \sigma \|u_1 - u_2\|_\infty, \end{aligned}$$

for some $\sigma \in (0, 1)$, provided ε is chosen small enough.

Clearly, the above inequality immediately implies that there exists a unique element $\gamma_* \in \Gamma_{\gamma,c}$ such that $\hat{T}\gamma_* = \gamma_*$, this is the *local* unstable manifold of 0.

Regularity—a cone field

As already mentioned, a separate argument it is needed to prove that γ_* is indeed a $\mathcal{C}^{(1)}$ curve.

To prove this, one possibility would be to redo the previous fixed point argument trying to prove contraction in $\mathcal{C}_{Lip}^{(1)}$ (the $\mathcal{C}^{(1)}$ functions with Lipschitz derivative); yet this would require to increase the regularity requirements on T . A more geometrical, more instructive and more inspiring approach is the following.

Define the cone field $\mathcal{C}_{\theta,h}(x, u) := \{\xi \in B_h(x) \mid (a, b) = \xi - x; a \neq 0; |\frac{b}{a} - u| \leq \theta\}$, with $|u| \leq c\delta$, $\theta \leq c\delta$ and $h \leq \delta$. By construction $B_h(x) \cap \gamma_* \subset \mathcal{C}_{c\delta,h}$ for each $x \in \gamma_*$. We will study the evolution of such a cone field on γ_* .

For all $\xi \in \mathcal{C}_{\theta,h}(x, u)$, if $(a, b) = \xi - x$ and $(\alpha, \beta) = T\xi - Tx$, it holds

$$(\alpha, \beta) = D_x T(a, b) + \mathcal{O}(C\|(a, b)\|^2).$$

Thus, setting $(\alpha', \beta') = D_x T(a, b)$ and $u' = \frac{\beta'}{\alpha'}$,

$$\left| \frac{\beta}{\alpha} - u' \right| \leq \mu\lambda^{-1}[c_1 h + \theta],$$

for some constant c_1 depending only on T and δ . Accordingly, if h is small enough, there exists $\sigma \in (0, 1)$ such that

$$T\mathcal{C}_{\theta,h}(x, u) \subset \mathcal{C}_{\sigma\theta,h}(Tx, u').$$

Hence, if $x \in \gamma_*$, $\gamma_* \cap B_{\sigma^{-n}h}(T^{-n}x) \subset \mathcal{C}_{c\delta,h}(T^{-n}x, 0)$ and, since $T^{-n}\gamma_* \subset \gamma_*$,

$$\gamma_* \cap B_h(x) \subset \mathcal{C}_{\sigma^n c, h}(x, v_n) \quad (3.3.8)$$

where $(a, av_n) = D_{T^{-n}x} T^n(1, 0)$ and $h < \sigma^n c\delta$.

The estimate (3.3.8) clearly implies

$$\gamma'_*(x) = (1, \lim_{n \rightarrow \infty} v_n) \quad (3.3.9)$$

which indeed exists (see Problem 3.9). $\square$

There is an issue not completely addresses in our formulation of Hadamard-Perron theorem: the uniqueness of the manifolds.⁷ It is not hard to prove that $W^{s(u)}(p)$ are indeed unique (see Problem 3.10).

The point of view employed in the previous theorem is brought to its extreme consequences in [46].

There is another point of view that can be adopted in the study of stable and unstable manifolds: to “grow” the manifolds. This is done by starting with a very short curve in $\Gamma_{\delta,c}$, e.g. $\gamma_0(t) = (t, 0)$ for $t \in [\lambda^{-n}\delta, \lambda^n\delta]$, and showing that the sequence $\gamma_n := T^n\gamma_0$ converges to a curve in the strip $[-\delta, \delta]$, independent of γ_0 . From a mathematical point of view, in the present case,

⁷Namely the doubt may remain that a less regular set satisfying Definition 3.3.1 exists.

it corresponds to spell out explicitly the proof of the fixed point theorem. Nevertheless, as we will see in ???, it is a more suggestive point of view and it is more convenient when the hyperbolicity is non uniform. For example consider the map⁸

$$T \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} 2x - \sin x + y \\ x - \sin x + y \end{pmatrix} \quad (3.3.10)$$

then 0 is a fixed point of the map but

$$D_0 T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not hyperbolic, yet, due to the higher order terms, there exist stable and unstable manifolds (see Problems 3.11, 3.12, 3.13).

3.4 Invariant manifolds and foliations (existence)

The concept of stable and unstable manifold of a fixed point can be obviously generalized to periodic orbits and, less obviously but more interestingly, to any orbit:

Definition 3.4.1 For each point $x \in X$ we define its stable (unstable) sets as

$$W^{s(u)}(x) = \{z \in X \mid \lim_{n \rightarrow \infty(-\infty)} d(T^n x, T^n z) = 0\}.$$

Remark 3.4.2 It will not surprise the reader that very often such sets are indeed manifolds, in this case they are called stable (unstable) manifolds. In the following we will discuss only this case.

Very often the structure of the stable manifolds is quite complex (as we have already seen in the introduction) and for this reason it may be more convenient to talk about *local* stable (unstable) manifolds as we did already (see Definition 3.3.1).

Definition 3.4.3 By local stable (unstable) manifolds at $x \in X$ we mean the connected component containing x of $B_\varepsilon(x) \cup W^{s(u)}(x)$, for some $\varepsilon > 0$.

Clearly the local manifolds are never unique, since their size may vary, but they may be unique once the size is assigned. Sometimes local stable manifolds will be denoted by $W_{\text{loc}}^s(x)$ as well as $W_\delta^s(x)$, we will use the former only if some confusion can arise and the latter only if the size plays an explicit rôle in the argument.

⁸Some times this is called *Levowich map*.